Image Generation using (β-)VAE, VQ-VAE and WGAN

Assignment Overview

In this assignment, you will implement and compare generative modelling approaches for images:

1. **(β-)Variational Autoencoder (β-VAE)**
2. **Vector Quantized VAE (VQ-VAE)** with **PixelCNN** prior for latent sampling
3. **Wesserstein Generative Adversarial Network (WGAN)** and **WGAN-GP**

The goal is to understand how different techniques influence **reconstruction quality**, and **sample realism** in images. Use provisioned WILP lab infrastructure (mandatory)!

Dataset

Use **CIFAR 10** dataset.

Preprocessing

- Normalize pixel values to [0,1]

Part A:

(β-)VAE [2 Marks]

Use $\beta \in \{1, 2, 4, 10\}$

Tasks

1. Train models with different β values
2. Compare reconstruction quality (PSNR) w.r.t. epochs for different β values
3. Compare FID scores for different β values

VQ-VAE with PixelCNN Prior [3 Marks]

Use Codebook size K (512, 256, 128)

Tasks

1. Train VQ-VAE and report reconstruction quality
2. Extract discrete latent maps
3. Train PixelCNN over latents
4. Sample new latent codes using PixelCNN
5. Generate images from sampled latents

6. Compare FID scores, reconstruction quality (PSNR) and codebook quality for different K

Part B: WGAN and WGAN-GP [3 Marks]

Tasks

1. Train WGAN and WGAN-GP
2. Generate random samples
3. Compute FID scores

Part C: Comparative Analysis [2 Mark]

Include 100 visual examples for VAE (β=1), VQ-VAE (K=256), WGAN and WGAN-GP and comment on perceptual quality.

Compare evaluation scores for (β-)VAE, VQ-VAE, WGAN and WGAN-GP

Deliverables

1. **ipynb** notebooks (PyTorch / TensorFlow) for Part A, and Part B
2. Corresponding html files
3. Pdf file for Part C including comparative discussion (reconstruction error, sampling quality, speed)

Marks will be deducted for unsatisfactory values of evaluation metrics.

Submission Instructions

- Upload both *.ipynb with all relevant outputs embedded and corresponding *.html files and the pdf file for the part C.
- GPU infrastructure in BITS Lab Infrastructure (Prayogshala) is made available for the UDL course (S1-26_AIMLZG533). Make use of this infrastructure and submit a screenshot in jpg format showing usage of BITS provided infrastructure for this assignment.
- Marks will be deducted for inadequate training resulting and higher or unexpected results in all tasks! Your results should validate the well-established characteristics of different architectures for the provided datasets.
- Note that this assignment is a group assignment and the same assignment1 grouping will be followed for this assignment2 as well. As it is a group assignment, one submission on behalf of the group will be enough.